

Problema 1.22

$$\theta_n = \frac{2\pi}{2^n}$$

$$p_n = 2^n \frac{1}{2} \sin \theta_n = 2^{n-1} \sin \theta_n$$

$$\begin{aligned} \lim_{n \rightarrow \infty} p_n &= \lim_{n \rightarrow \infty} 2^{n-1} \sin \frac{2\pi}{2^n} = \lim_{n \rightarrow \infty} \frac{\sin(2^{1-n}\pi)}{2^{1-n}} \stackrel{l'H\hat{o}spital}{=} \lim_{n \rightarrow \infty} \frac{\pi \cos(2^{1-n}\pi) \ln 2 (-2^{1-n})}{\ln 2 (-2^{1-n})} \\ &= \lim_{n \rightarrow \infty} \pi \cos \left(\underbrace{2^{1-n}\pi}_{\xrightarrow[n \rightarrow \infty]{0}} \right) = \pi \cos(0) = \pi \end{aligned}$$

$$\begin{aligned} \frac{\sin \theta_{n-1}}{\left[2\left(1 + \sqrt{1 - \sin^2 \theta_{n-1}}\right)\right]^{\frac{1}{2}}} &= \frac{\sin \theta_{n-1}}{\left[2\left(1 + \sqrt{\cos^2 \theta_{n-1}}\right)\right]^{\frac{1}{2}}} = \frac{\sin \theta_{n-1}}{\sqrt{2 + 2 \cos \theta_{n-1}}} = \frac{1}{\sqrt{2}} \sqrt{\frac{\sin^2 \theta_{n-1}}{1 + \cos \theta_{n-1}}} \\ &= \frac{1}{\sqrt{2}} \sqrt{\frac{1 - \cos^2 \theta_{n-1}}{1 + \cos \theta_{n-1}}} = \frac{1}{\sqrt{2}} \sqrt{1 - \cos \theta_{n-1}} = \frac{1}{\sqrt{2}} \sqrt{2 \sin^2 \frac{\theta_{n-1}}{2}} = \sin \frac{\theta_{n-1}}{2} = \sin \theta_n \end{aligned}$$